

The most famous of the park's mammals are the bats. The park hosts 17 different bat species. The large colony of Brazilian free-tailed bats wow visitors every evening from spring through fall with its spectacular outflights. Two other species have also been found regularly in Carlsbad Cavern—cave myotis and fringed myotis bats. They typically roost in a different part of the cavern and fly about 1.5 miles (2.4 km) before exiting the Natural Entrance.

But not all bat species roost in caves. Among the other species using the park are eastern red bats and hoary bats, which roost in trees, and parastrellus hesperus or canyon bats, which roost on rock cliffs and in cracks. Bats are mammals, which means that they give live birth to their young (do not lay eggs), are warm-blooded, have fur (not feathers), and baby bats or pups are fed breast milk (not insects) by their mothers. Bats are the only true flying mammals. All the bats in the area around Carlsbad Caverns National Park are insectivores. The Brazilian free-tailed bats weigh about 1/2 ounce (13 g), which is equivalent to the weight of three nickel coins. Their wingspan is approximately 11 inches (28 cm). Bat numbers in the Cavern are variable. The resident colony was around 400,000 in summer of 2005. During the spring and fall migration, the bat numbers in the cavern were documented as high as 793,000 in 2005. There are seasonal fluctuations of the numbers, as well as daily fluctuations. Researchers from Boston University have been assisting the park in getting accurate population estimates. They use advanced thermal infrared imaging cameras coupled with a custom-written visual recognition software program to count the bats. At Carlsbad Cavern, the resident colony should not be called the maternity colony because it is typically greater than 50% male. The males and females roost mix together in the same site. In many sites outside of the park, the really large Brazilian free-tailed bat colonies are almost exclusively female and the males roost separately in smaller groups. The health and well-being of bat colonies all over the United States is threatened by a deadly fungus affecting bats called White-nose Syndrome (WNS). WNS has already taken the lives of an estimated 7+ million bats in the United States and expected to spread wider than it is today. The bats that call the cavern home in the summertime have not been affected by this fatal fungus, and we do everything we can to keep it that way. We will ask you questions when you arrive to ensure you do not bring the fungus with you from another cave. Although this fungus is not harmful to humans, it can spread quickly through a bat colony with disastrous results. Learn more about White-nose Syndrome.

## Benefits of Bats

Sure, it's interesting that bats navigate by echolocation and that they're nocturnal. But do they really matter? The short answer is "Yes!" These flying mammals bring many benefits to their ecosystems. More than 50 unique species of bats live in national parks, and different species provide different benefits. Some pollinate plants, others eat insects, many serve as prey to other animals, and they all inspire scientific discoveries.

## Some Bats are Pollinators

Several species of bats in tropical and subtropical areas of the Americas eat nectar. Many types of plants in these regions rely on bats for pollination and seed dispersal, such as the blue agave. In some southwestern parks, long-nose and long-tongue bats are perfectly adapted to pollinate these plants, and they provide extensive value to the agricultural industry. So next time you sweeten your coffee with agave nectar, remember to thank a bat.

## Bat Guano Supports Cave Communities

Caves are complex and unique ecosystems that provide homes for a diversity of creatures from insects to amphibians and fish as well as mammals like wood rats and bats. Many of these creatures can only survive within the cave, and they rely on nutrients carried into the cave by water or other animals. Bats benefit caves by providing important nutrients in their guano (better fertilizer than cow manure!) that support the growth of communities of cave organisms.

### Some Bats Provide Insect Control

Bats that eat insects are called "insectivorous." They feast on insects each night, adding up to more than \$3.7 billion worth of pest control each year in the U.S. When bats are around to eat insects, there are fewer insect pests causing damage to crops, and farmers don't have to invest as much in pesticides. Bats can eat 50% or more of their bodyweight in insects each night. Imagine eating 200 quarter-pound burgers—that's how much a bat eats in insects in one night!

### Bats Are Also Prey

Just as some bats rely on thousands of insects each night for survival, other animals in the ecosystem rely on bats for their calories. Hawks, falcons, and owls eat bats, and mammals like weasels, ringtail cats, and raccoons sometimes attack bats while they roost.

### Fruit-Eating Bats Disperse Seeds

Fruit-eating bats play important roles in distributing seeds to maintain plants and forests. These species of bats, often called "flying foxes" because of their larger body size and big eyes, live in tropical and subtropical areas of the Old World (Africa, Asia and Australia). Fruit-eating bats are also found in some Pacific islands, Latin America, and the Caribbean and live in national parks in Guam, American Samoa, and the Virgin Islands!

### Bats Provide Inspiration

Some of bats' unique features like membrane wings and echolocation have inspired technological advances in engineering. Drones that have thin and flexible bat-like wings are in the works as well as tiny, more efficient sonar systems for navigation. The wingsuits used by basejumpers take more than a few cues from bats' aerodynamic bodies.

### White-nose Syndrome

White-nose syndrome (WNS) is a fatal disease of hibernating bats caused by the fungus *Pseudogymnoascus destructans* (Pd for short). The fungus infects bats while they are hibernating over winter. Researchers first discovered the disease in New York in the winter of 2006. Since then, it has spread across much of the United States and Canada, killing millions of bats—up to 99% of some bat colonies. The National Park Service works with many other state and federal agencies, tribes, universities, and conservation organizations who are part of the White-nose Syndrome Response Team to learn more about the fatal disease and how to reduce impacts to bats.

### Help Bats

We need bats if we want healthy and diverse ecosystems filled with a variety of organisms. Many bats feast on insects, some pollinate plants, and some bats spread seeds, too. Larger animals may depend on bats for prey. Aside from all these great services, bats are the only mammal that's capable of true flight, and they're simply fascinating. The deadly fungal

disease, white-nose syndrome (WNS), poses a real danger to cave-dwelling bats. WNS has killed 99% of some bat colonies. Tree-roosting and migratory bats face significant threats from wind turbines. Habitat loss and climate change also make it difficult for bats to survive and thrive. But we can all help protect bats and their habitats! Try some of these ideas to help conserve bats. You can help bats in your own yard and community, and you can learn how to protect bats when you visit parks, too.

Stay out of caves when directed. The bats at risk for WNS often hibernate or raise young in caves. They need to be undisturbed so they can rest and raise their young.

Decontaminate before going in a cave. It may be possible for humans to spread WNS from one cave into another. If you have visited another cave, make sure to decontaminate (or leave behind) your shoes, clothes, and gear before entering another cave.

Leave bats alone. If you see a bat, do not disturb it.

Tell a ranger if you see bats acting strangely. Rangers can take steps to protect both bats and people if a bat is behaving in an unusual way.